

Introduction to Software Engineering



Report Summary

Name	: Unathi Qamza
Your Score	: 38 out of 40 (95%)
Correct Answers	: 38 Questions
Incorrect Answers	: 2 Questions
Unanswered	: 0 Questions
Required Passing Grade (%)	: 50%
Time Taken	: 31 mins 14 secs
Date	: Mar 26, 2026
Your Result	: Pass
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Your Answers

Correct

- Q1) What is the main goal of software engineering?
- A. Writing code quickly
 - B. Designing hardware systems
 - C. Building reliable and efficient software (Your Answer)(Correct)
 - D. Debugging existing applications

Explanation:

The goal of software engineering is to design, develop, and maintain reliable and efficient software that meets user needs.

Correct

- Q2) Which of the following best describes the software development lifecycle (SDLC)?
- A. A model for website design
 - B. A process used to plan, develop, and maintain software (Your Answer)(Correct)
 - C. A method for testing hardware components
 - D. A way to teach programming

Explanation:

SDLC defines a structured process for planning, developing, testing, deploying, and maintaining software systems.

Correct

- Q3) Which SDLC phase involves determining the project's feasibility and scope?
- A. Maintenance
 - B. Design
 - C. Requirement analysis (Your Answer)(Correct)
 - D. Implementation

Explanation:

Requirement analysis identifies user needs, defines the system scope, and determines feasibility before design begins.

Incorrect

Q4) Which of the following is not a programming language?

- A. Python
- B. Java
- C. HTML (Your Answer)(Incorrect)
- D. Algorithm (Correct)

Explanation:

An algorithm is a step-by-step solution, not a programming language.

Correct

Q5) In programming, what does a variable do?

- A. Stores data temporarily (Your Answer)(Correct)
- B. Creates graphics
- C. Runs a loop automatically
- D. Encrypts information

Explanation:

A variable is used to store data values that can change during program execution.

Correct

Q6) Which one is an example of a software design pattern?

- A. Waterfall model
- B. Singleton (Your Answer)(Correct)
- C. Testing phase
- D. Agile

Explanation:

The Singleton pattern ensures a class has only one instance and provides a global point of access to it.

Correct

Q7) What does IDE stand for?

- A. Integrated Design Environment
- B. internal development engine
- C. Integrated Development Environment (Your Answer)(Correct)
- D. Internet Data Editor

Explanation:

IDEs like VS Code or IntelliJ provide tools for writing, debugging, and testing code efficiently.

Correct

Q8) Which model follows a linear sequential flow?

- A. Agile
- B. Spiral
- C. Waterfall (Your Answer)(Correct)
- D. RAD

Explanation:

The waterfall model is a traditional SDLC model where each phase flows linearly into the next.

Correct

Q9) During which SDLC phase is the actual code written?

- A. Design
- B. Implementation (Your Answer)(Correct)
- C. Testing
- D. Planning

Explanation:

Implementation (or development) is when programmers write and compile code based on the system design.

Correct

Q10) In agile development, work is divided into short cycles known as:

- A. Waterfalls
- B. Sprints (Your Answer)(Correct)
- C. Modules
- D. Phases

Explanation:

Agile uses sprints—short, iterative development cycles—to deliver functional software quickly.

Correct

Q11) Which of the following best describes encapsulation in programming?

- A. Reusing existing code
- B. Combining data and methods into a single unit (Your Answer)(Correct)
- C. Dividing a program into functions
- D. Inheriting attributes from another class

Explanation:

Encapsulation binds data and its functions together, protecting them from unauthorized access.

Correct

Q12) What is the main purpose of software architecture?

- A. To handle customer support
- B. To define the overall structure of a software system (Your Answer)(Correct)
- C. to test the program
- D. To write pseudocode

Explanation:

Software architecture defines how components interact, ensuring scalability, reliability, and performance.

Correct

Q13) Which programming concept focuses on objects and classes?

- A. Procedural programming
- B. Object-Oriented Programming (OOP) (Your Answer)(Correct)
- C. functional programming
- D. declarative programming

Explanation:

OOP organizes software design around objects that contain both data and behavior.

Correct

Q14) In the SDLC, what comes immediately after the design phase?

- A. Testing
- B. Deployment
- C. Implementation (Your Answer)(Correct)
- D. Planning

Explanation:

Implementation follows design, where developers write code to create the actual software product.

Correct

Q15) Which of the following ensures that software meets user requirements before deployment?

- A. Debugging
- B. Testing (Your Answer)(Correct)
- C. Deployment
- D. Design

Explanation:

Testing ensures software functionality, reliability, and quality before it's released to users.

Correct

Q16) A 'class' in object-oriented programming can be best described as:

- A. A variable type
- B. a data storage system
- C. A blueprint for creating objects (Your Answer)(Correct)
- D. A method library

Explanation:

A class defines the structure and behavior of objects, serving as their blueprint.

Correct

Q17) Which of the following is a non-functional requirement?

- A. User login
- B. Data encryption (Your Answer)(Correct)
- C. Payment feature
- D. User registration

Explanation:

Non-functional requirements define system qualities like security, performance, and usability.

Correct

Q18) What is the main goal of software design patterns?

- A. To increase software cost
- B. To solve recurring design problems efficiently (Your Answer)(Correct)
- C. To improve hardware performance
- D. To generate code automatically

Explanation:

Design patterns provide reusable solutions to common software design challenges.

Correct

Q19) Which SDLC model is best suited for projects with changing requirements?

- A. Waterfall
- B. Agile (Your Answer)(Correct)
- C. V-Model
- D. Incremental

Explanation:

Agile's flexibility allows teams to adapt quickly to evolving customer needs.

Correct

Q20) The process of finding and fixing bugs in software is called:

- A. Compilation
- B. Debugging (Your Answer)(Correct)
- C. Refactoring
- D. Designing

Explanation:

Debugging involves identifying and correcting errors to ensure the program runs correctly.

Correct

Q21) If a project team discovers a major design flaw after testing, which SDLC phase should they revisit?

- A. Design (Your Answer)(Correct)
- B. Implementation
- C. Planning
- D. Maintenance

Explanation:

Revisiting the design phase helps correct structural issues before further testing or deployment.

Correct

Q22) What is a key benefit of using modular programming?

- A. It increases redundancy
- B. It makes the code longer
- C. It enhances maintainability and reusability (Your Answer)(Correct)
- D. It reduces team collaboration

Explanation:

Modular programming breaks programs into reusable, manageable parts for easier maintenance.

Correct

Q23) Which design pattern would you use to manage communication between objects without creating tight coupling?

- A. Observer pattern (Your Answer)(Correct)
- B. Singleton pattern
- C. Factory pattern
- D. Builder pattern

Explanation:

The observer pattern lets multiple objects listen and respond to changes in another object without direct dependence.

Correct

Q24) What is the primary advantage of the agile model over the waterfall model?

- A. Predictable project cost
- B. Fixed requirements
- C. Continuous feedback and adaptation (Your Answer)(Correct)
- D. Reduced client involvement

Explanation:

Agile promotes flexibility and iterative improvements through regular client feedback.

Correct

Q25) Which SDLC phase ensures that the software operates correctly after release?

- A. Testing
- B. Deployment
- C. Maintenance (Your Answer)(Correct)
- D. Requirement analysis

Explanation:

The maintenance phase involves bug fixes, updates, and performance improvements after deployment.

Correct

Q26) What does 'abstraction' mean in object-oriented programming?

- A. Hiding complex details and showing only essential features (Your Answer)(Correct)
- B. Copying methods between classes
- C. Combining unrelated data
- D. Displaying all class details

Explanation:

Abstraction simplifies complexity by focusing only on key details relevant to the user or system.

Correct

Q27) Which of the following best represents a layered software architecture?

- A. User interface → business logic → database (Your Answer)(Correct)
- B. Database → business logic → user interface
- C. Hardware → operating system → user
- D. Compiler → runtime → debugger

Explanation:

Layered architecture separates concerns, improving scalability and maintainability.

Correct

Q28) Which design pattern is most useful for creating families of related objects without specifying their concrete classes?

- A. Singleton
- B. Abstract factory (Your Answer)(Correct)
- C. Prototype
- D. Decorator

Explanation:

The abstract factory pattern provides an interface for creating related objects without knowing their exact classes.

Correct

Q29) A software engineer is designing an e-commerce system. To reuse code for similar products, which concept should they apply?

- A. Inheritance (Your Answer)(Correct)
- B. Encapsulation
- C. Composition
- D. Polymorphism

Explanation:

Inheritance enables code reusability by allowing new classes to adopt properties from existing ones.

Correct

Q30) Which principle of software design encourages “loose coupling” and “high cohesion”?

- A. SOLID principles (Your Answer)(Correct)

- B. Agile manifesto
- C. SDLC phases
- D. UML diagrams

Explanation:

SOLID principles promote modular, maintainable, and flexible software design through loose coupling and high cohesion.

Correct

- Q31) Which principle emphasizes the importance of a class having only one reason to change?
- A. Single Responsibility Principle (Your Answer)(Correct)
 - B. Open/Closed Principle
 - C. Liskov Substitution Principle
 - D. Interface Segregation Principle

Explanation:

The Single Responsibility Principle states that a class should have only one reason to change.

Correct

- Q32) What principle states that software entities should be open for extension but closed for modification?
- A. Single Responsibility Principle
 - B. Open/Closed Principle (Your Answer)(Correct)
 - C. Liskov Substitution Principle
 - D. Dependency Inversion Principle

Explanation:

The Open/Closed Principle states that software entities should be open for extension but closed for modification.

Correct

Q33) Which principle suggests that derived classes must be substitutable for their base classes?

- A. Single Responsibility Principle
- B. Open/Closed Principle
- C. Liskov Substitution Principle (Your Answer)(Correct)
- D. Interface Segregation Principle

Explanation:

The Liskov Substitution Principle suggests that derived classes must be substitutable for their base classes.

Correct

Q34) What principle advocates for creating small, specific interfaces rather than large, general-purpose ones?

- A. Single Responsibility Principle
- B. Open/Closed Principle
- C. Liskov Substitution Principle
- D. Interface Segregation Principle (Your Answer)(Correct)

Explanation:

The Interface Segregation Principle advocates for creating small, specific interfaces rather than large, general-purpose ones.

Correct

Q35) Which principle emphasizes the importance of depending on abstractions rather than concrete implementations?

- A. Single Responsibility Principle
- B. Open/Closed Principle
- C. Dependency Inversion Principle (Your Answer)(Correct)
- D. Interface Segregation Principle

Explanation:

The Dependency Inversion Principle emphasizes the importance of depending on abstractions rather than concrete implementations.

Correct

Q36) What is the main goal of the SOLID principles?

- A. To create complex software
- B. To promote software design best practices (Your Answer)(Correct)
- C. To eliminate all bugs
- D. To ensure software is free of documentation

Explanation:

The main goal of the SOLID principles is to promote software design best practices.

Correct

Q37) Which of the following is NOT one of the SOLID principles?

- A. Single Responsibility Principle
- B. Open/Closed Principle
- C. Liskov Substitution Principle
- D. Agile Manifesto (Your Answer)(Correct)

Explanation:

The Agile Manifesto is not one of the SOLID principles.

Correct

Q38) A company needs a scalable system that can integrate new features easily. Which architectural pattern suits this best?

- A. Monolithic
- B. Layered
- C. Microservices (Your Answer)(Correct)
- D. Waterfall

Explanation:

Microservices architecture promotes scalability by dividing applications into small, independent services.

Incorrect

- Q39) When designing a system that must handle continuous user input and real-time responses, which design consideration is most important?
- A. Static data
 - B. Performance and responsiveness (Correct)
 - C. Offline functionality
 - D. Manual updates (Your Answer)(Incorrect)

Explanation:

Real-time systems prioritize performance and responsiveness to ensure smooth user experiences.

Correct

- Q40) A developer modifies existing code without changing its external behavior to improve readability. This process is called:
- A. Debugging
 - B. Refactoring (Your Answer)(Correct)
 - C. Recompiling
 - D. Unit testing

Explanation:

Refactoring restructures code internally for clarity and maintainability while preserving functionality.